A COMPARATIVE ANALYSIS OF MACHINE LEARNING CLASSIFIERS FOR DETECTING FAKE NEWS

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SUPERVISOR’S APPROVAL

The thesis titled “A Comparative Analysis of Machine Learning Classifiers for detecting Fake News” Submitted by Md Fayek Mubarrat Jamee (Roll No.: 150129), Minhaz Uddin Sadif (Roll No.: 150159), Abu Yusuf Patwary (Roll No.: 150109), Session: 2014-15 has been accepted as satisfactory in partial fulfillment of the requirement for the degree of Bachelor of Science in Information and Communication Engineering.

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DECLARATION

We hereby declare that this thesis is our original work and it has been written by us in its entirety. We have duly acknowledged all the sources of information which have been used in the thesis. The thesis (fully or partially) has not been submitted for any degree or diploma in any university or institute previously.

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ABSTRACT

This thesis work compares the accuracy of different machine learning classification algorithms to detect a fake news. A fake news can be termed as false stories, statements or opinion disguised as news mainly spread in the internet or other media, usually created with an intent of manipulating mass people's views on a specific subject, especially political matters, damage someone's or an organization's reputation or financial position or simply as a joke. We used a dataset obtained from politifact.com and BuzzFeed and combined them together. The result was a dataset of 701 news. We apply Term Frequency, Inverse Document Frequency (TF-IDF) to a corpus created from this dataset. We apply multiple classification algorithms like Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, Neural Networking, Naive Bayes (Bernoulli) and K Nearest Neighbor. Then we test the algorithms and compare them based on their accuracy, precision, recall and f1-score. We found that Support Vector Machine has the highest accuracy, Logistic Regression has the highest precision. On the other hand, for recall value K Nearest Neighbor has the highest values and Support Vector Machine had the highest f1 score.

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CHAPTER ONE

INTRODUCTION

1.1 Motivation

The spreading of fake news has been a matter of concern for us from the beginning of human civilization. Whether it was to damage someone's reputation, popularity, or to manipulate people's choices in an election or even people's buying choices, spreading fake news have always been a devious weapon for people in wrongdoings. It is even more devastating in the age of internet and social media. People will spread rumors, false claims, and misleading information just to gain popularity, more likes, shares, clicks and page views. As an example, [1] showed the important impact of fake news in the context of the 2016 US presidential elections. Many online news portals even make this a business strategy to spread enticing or click-bait news headlines and articles to get more clicks so they get 'viral' and become more popular. In this report, we try to create a model that, if given a news article, can accurately calculate the likelihood of that article being real or fake. A research [7] analyzed the most viral tweets related to the Boston Marathon blasts in 2013, finding that the number of fake tweets is higher than number of real.
With the modern age of information, we are at a peak of possibilities with all the information we have at our disposal. Anyone can ask for information about anything and freely get it from various online information storage. Any new information or news spreads faster than a wildfire reaching the other end of the worlds within seconds if not less. But with all these benefits also comes a possibility of danger. Wrong information, misleading information, manipulated facts, half-truths and lies can be very devastating in our society. It holds the power to destroy any people or organization socially and financially.

Tech giants like Google and Facebook have been trying to tackle the fake news issues recently. Facebook has recently given its users an option to flag news as fake or misleading. But they are trying to automate the process so that their system automatically detects a possible fake or misleading news and take necessary action. It is a particularly tough and sensitive task. Because the algorithm that is used to detect a fake news should be politically unbiased. It is because fake news can come from different political ends to facilitate their agenda. It is also important to determine which sources should be considered authentic and based on which features

1.2 Objectives and Goals

The main focus of this paper is to compare various machine learning classification algorithms like logistic regression, Naive Bayes, support vector machine, decision tree, random forest, neural networking and K nearest neighbor for detecting fake news based on four different factors: accuracy, precision, recall and f1 score.

1.3 Brief Working Procedure

In this paper, we compare the performance of models with different machine learning algorithm based on TF-IDF bi-gram frequency. We use TF-IDF and not other methods because TF-IDF bi-gram creates predictive models that are very effective for the classification of articles that are from an unknown or unreliable source.

For this, we used a dataset from politifact.com and BuzzFeed. We combine the data found in those 2 datasets. The result is a list of 701 news article categorized as Real or Fake. In our dataset, we have 203 Real news and 498 fake news. We split the dataset into 2 portions, one of which is used to train the data, while the other portion is used to test the algorithms. The first portion consists of 80% of the whole dataset and the second portion is 20% of the dataset.

After that, we feed the training data to different machine learning classification algorithms: Logistic Regression, Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, Neural Networking, Naive Bayes (Bernoulli) and K Nearest Neighbor. For each algorithm, we then detect the accuracy for the particular algorithm and the confusion matrix of them using the test portion of the dataset. We take the best accuracy when the algorithm provides different accuracy for different parameters, like C in the case of Support Vector Machine.

1.4 Paper Outline

In Chapter 1 Introduction, Motivation, Objective and Goals are described. Chapter 2 briefly describes the past works done in the field of using text classification for detecting fake news. Chapter 2 also includes different types of fake news that are seen on the internet. Chapter 3 looks at the machine learning classification we are using and their working principles and compares them based on theoretical analysis. Chapter 4 looks at the experimental results we got by

four factors, accuracy, precision, recall and f1 score.

Chapter 5 presents the conclusion and briefly discusses the scope of further improvement in this line and the proposed methodology.

CHAPTER TWO

REVIEW OF LITERATURE

2.1 Related Works

Approaches to fake news detection is not a new thing in the research area. Global research is going throughout the world. Applying the machine learning and natural language processing algorithm is also not new. In this, we have tried to decrease the fake news possibility to some extent. However, the possibility of detecting the fake news is very limited due to properly trained dataset and the big amount of data on the internet.

The World Wide Web has changed the dynamics of information transmission as well as the agenda-setting process [2]. Importance of facts, in particular when related to social relevant issues, mix with half-truths and false information to create informational combinations [3, 4]. In such a situation, as pointed out by [5], people can be uninformed or misinformed and the role of corrections in the diffusion and formation of biased beliefs are not effective. In specific, in [6] online campaigns have been shown to create a strengthening effect in usual consumers of conspiracy stories. Most of the time, Fake news is created intentionally to create doubt to the reader's mind. Fake news comes with the form of the real news because the creator of the news intentionally tries to prove the wrong information right. This is one of the most challenging parts of detecting the fake news. Research has shown that many times there is some misconception created by misleading titles. On that news title news and the body of the news totally different. Fake news is also involved with human emotions. Sentiment and conceptions. Most of the time it is hard to decide if the news is a joke or true news. Some of the times motives and thoughts can be varied by person to person. If someone believes are related to certain news which it thinks is true but many other people think it is false. It is hard to decide if the news is true or fake. These are some of the challenges of detecting fake news. The prediction of the fake news is very critical because it involved with the public emotion and sentiment. Also, it is hard to detect the motive of the news presenter with some defined datasets because the resources are huge. For this reason, two types of fake news can be found on the internet. One is knowledge-based another one is style based. It is very hard to detect the knowledge-based fake news. Because we cannot just put all the pieces of knowledge in a single system and find fake news from that. Also, the news is changing every day and every time. That's why in this paper we have tried to focus on the style based on fake news detection. Which detects the fake news from authors writing style or writing approach.

The main purpose of this thesis work is to apply machine learning algorithms to detect pattern of the fake news. The present scope of this research paper is huge. Big companies like Google, Facebook are also doing research in this field and more importantly the world is concern about the bad impact of Fake news detection. Recently in Bangladesh, there were a huge clash happened and many of students got injured. The whole incident happened due to spreading of the fake news and not correcting it on the spot. So, by this research paper we have tried to avoid this fake news by using some parameters.

The old way of verifying online content, via "manual" knowledge-based fact-checking [8], is made difficult – or, practically, impossible – by the "huge size of information that is now generated online" [9]. Therefore, various authors decide on the need of automatic, computational hoax detection systems [10], [11], [12]. Fake news detection methods have been classified into two types - news content models and social context models - based on their main input sources [8]. In news content-based approaches, we detect news based on the content of the news, headline, body, images [14]. Social Context means the reaction or sentiment of the mass people who are using social media on the internet. Therefore, analyzing [13], we can say there are mainly two ways to detect a fake news, fact checking and linguistic pattern.

But this is so alarming concern to the world but the related works are not too much up to the mark. The problem is bigger but the solves are not sufficient enough. Some of the work that has created a benchmark in this field in Liar Liar Pants on fire by William Yang Wang[15]. In this research paper, they have classified their models into six different scales. Those are True, Mostly True, Half True, Mostly False, False, Pants on fire. When the statement is accurate and there's nothing significant is missing then it is True. If the statement is true but needed correction then it is Mostly true. If the news is particularly true but creates misleading information then it is half true. There are 3 different scales for fake news as well. Fake News linguistic-based, according to

Liar Liar Pants on Fire research paper: There are three confusing article types here [15]. Satire is mimicking the real news and the reader will know it's not true. It is done for an only fun purpose. No desire to mislead people here. Hoax means convincing the readers to believe something according to their own perceptions convinces readers. Propaganda is used for mislead the readers. It is used to promote political issues. They classified these as a fake news.

They have used 12.8K datasets in [15] to implement their model with Politifact.com, where they have the justification for each of the false news check. We can see this graph from [15],

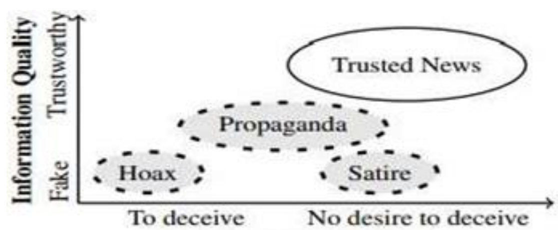


Figure 2.1: Intent of Author [15]

Another research work, Automatic Detection of Fake News [13],
This paper separates the task of fake news detection into three, by type of fake:

a. Serious fabrications (uncovered in mainstream or participant media, yellow press or tabloids)
b. Large-scale hoaxes,
c. Numerous fakes (news satire, parody, game shows). Serious fabricated news may take substantial efforts to collect, case by case. Journalistic fraudsters may face harsh consequences for dishonest reporting.

They did collect the dataset from computer-mediated communication (CMC), corpora (of e-mails, forum posts, etc.) are collected via elicitations or submissions of pre-existing messages. In NLP (or text analytics), other types of fake data have been collected by crawling the web or crowdsourcing.

The paper has some Requirements for Fake News Detection Corpus.

a. Availability of both truthful and deceptive instances
b. Digital textual format accessibility
c. Verifiability of “ground truth”.
d. Homogeneity in lengths.
e. Homogeneity in writing matter.
f. Predefined timeframe.
g. The manner of news delivery

Depending on these requirements they did separates fake news into three types. Finally, they decide, Three Types of Fake News Form Three Sub-Tasks in Fake News Detection:

a. Exposed fabrications (Shingler, 2015)
b. Large-scale hoaxes (Matt, 2015)
c. News satire (Fan Duel, 2015)

From this chapter we have learned about the different research works in this topic and we have found some outstanding research work is going on throughout the world. There are mainly two types of way to detect fake news, News content based and social media based. And that can be done either by fact checking or by knowledge-based analysis.

In the next chapter, we will learn about different classifiers which are used in the implementation for the analysis.

CHAPTER THREE

THEORETICAL ANALYSIS

3.1 Overview

In this chapter we are going to take a theoretical look into the process. First different processes to detect a fake news. Then we will discuss why we are using linguistic pattern analysis. After that we will take a look into the machine learning classifier algorithms that we are going to use and how they work. Finally, we will discuss the four comparison factors that we are going to use to compare the classifier algorithms.

3.2 Different ways to detect fake news

Different approaches can be taken to detect a fake news. We can categorize the various approaches into two main classes.

a. News Content Based: In these types of approaches, we analyze the content of the news to detect if it is real or fake.
b. Social Context Based: These types of approaches are suitable for news spread in social media. It mainly checks public responses on specific news to detect the acceptability of the news.

We are going to use the news content-based approach in this paper. It is because public reaction to the news on social media cannot always tell the truth. It is especially true for fabricated news that is specifically made to attract public sentiment and go viral. Recently we have seen many news going ‘viral’ on social media which are actually fake.

There are mainly two kinds of approaches in News Content Based fake news detection.

a. Knowledge Based: In this method, we check the unknown news, and try to cross-reference or match it with existing news from a source that we already know is reliable. This is also known as fact-checking.
b. Linguistic Pattern Based: Fake news writers often have malicious intent, intent to miss lead or miss direct people or intent to dishonor any person or organization. These require some specific writing pattern that doesn't always appear in real news. This method of finding a specific pattern that is common in our dataset of fake news and analyzing unreliable news based on that pattern is linguistic pattern based fake news detection.

In this thesis work, we are going to use Linguistic Pattern Based approach to detect a fake news. In the next section, we are going to discuss the algorithms we will be using in our comparison and then look at a theoretical comparison between them.

3.3 Different Classification Algorithms

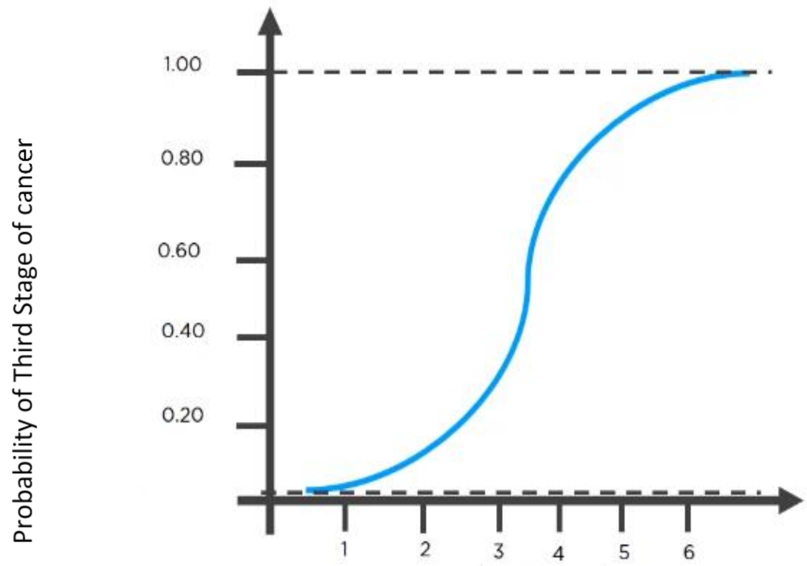
We have followed six different machine learning classification algorithms. Those are Logistic Regression, Support Vector Machine, Decision Trees, Random Forrest, K Nearest Neighbor and Naïve Bayes (Bernoulli). We are briefly going to discuss about these classifiers in this section.

3.3.1 Logistic Regression

Logistic regression is an algorithm for performing binary classification. So, the outcome should be discrete or categorical such as 0 or 1, True or False, Yes or No.

To understand the logistic regression, we can consider an example. A case is assumed that, a Doctor wants accurately diagnose Blood cancer. This disease is very much dangerous and often people die from this. The doctor wants to calculate the how long his second stage cancer patient

can survive. To determine this, previous patient data can be used. Here the regression model is created by using the information of other patients.



Years since suffering from Cancer

Figure 3.1: Logistic Regression

From the figure above, we can say that as the number of years increases the chances of affected people to get more affected is also increases.

In this classification, the predicted outcomes will come up for a given set of independent variables whereas dependent variable outcome is discrete.

For logistic regression a threshold value should be taken. For the above figure if we take threshold value, .5, when probability will be greater than .5 it will be rounded off to one.

The equation of logistic regression is given,

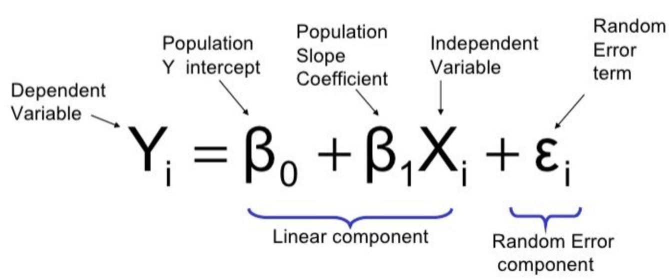


Figure 3.2: Explanation of Logistic Regression equation

The benefits of using Logistic Regression are,

a. It gives good information about the features.
b. It can be implemented in any type of data sets, small or big.
c. It is robust cause it does not require variables to be normally distributed.

The problem of logistic regression is it fall short if the decision boundary is nonlinear. It does not require normal distribution, so distribution can have little overlap.

Logistic Regression can be used in,

a. Weather Prediction.
b. Determining illness with logistic repression.
c. Image Identification.

There are two types of Regression, Logistic & Linear regression. Logistic regression is different from linear regression,

Table 3.1: Linear Regression vs Logistic Regression

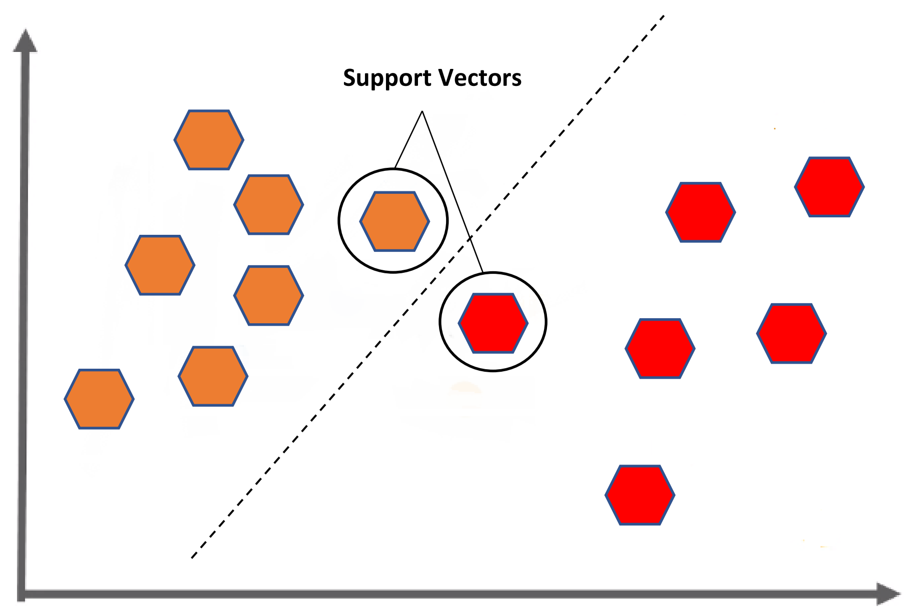
Linear Regression	**Logistic Regression**
It is used to solve regression related problems	Logistic Regression is used to classification problem
Linear regression is straight-line.	Logistic Regression curve is sigmoid.
The variables are continuous.	The variables are categorical.

Logistic regression can be very useful classifier because it is great when it comes to extracting a features information. Next, we will discuss about Support Vector Machine (SVM).

3.3.2 Support Vector Machine

A simple way to classify observation into two classes is to draw a line between them. If the data is perfectly separable in this way there are many possible boundaries that could be made. How can we understand which is the best? One approach is to use some kind of statistical distribution. This means the point which are far from the boundary must have an influence on them. It is analyzed that boundary should be made as far as possible from any of the observations. This is basic idea behind Support Vector Machine classification.

If the data is linear separable, this results in simple optimization problem. We have to choose the co-efficient of the linear boundaries to maximize the margin which is distance between the boundary and the nearest observations. In the end the optimal solution is determined only by the observation nearest to the boundary. These observations are referred to support vectors.



Predictor Variable

Figure 3.3: Support Vector Machine

Real noisy data may not be separable that is there is no linear variable that can classify every observation. In this case, we can modify the optimization the maximum the margin but with a penalty term for misclassified observations. Observations are correctly classified only if we put correct side observation in one side and wrong side in another. The penalty term creates a solution by creating a huge margin. SVM solution than gives the best possible separable solution between the classes that is the widest margin without unnecessary misclassification.

The advantages of SVM are,

a. Support vector machines are useful for high dimensional spaces.
b. SVM is memory efficient because it uses subset of training factors as support vector or decisive vector for classification
c. SVM have a very few irrelevant features.

The accuracy of SVM is low when the dataset is huge. When the dataset has more noise, it does not perform well.

Application of SVM is Face Detection, Hand writing recognition, Classification of images etc.

In the next part we will learn about Decision Tree Learning.

3.3.3 Decision Trees

Decision tree is one of most classic approaches to divide an information or data. In classification datasets are divided into different categories and labeled based upon their characteristics. Decision tree is a tree shaped graphical representation used to determine the possible solution. Decision tree contains branches like the actual tree. Here each branch represents a possible decision, based on some conditions.

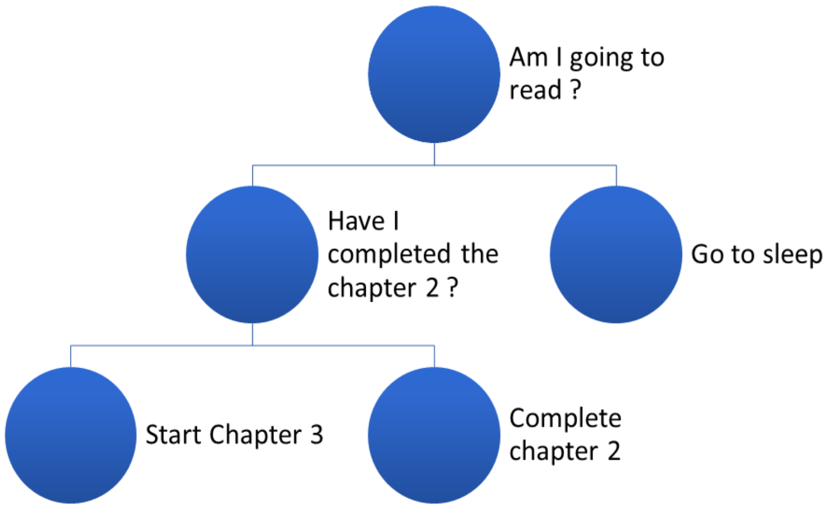


Figure 3.4: Decision Tree

In the figure above, we have plotted a decision tree. Here each branch referring to a possible decision. At first, we are staring by asking a simple question, Am I going to read? If it is No then it will make a decision go to sleep. If it is Yes, then second question is asked. Similarly, we can create lots of decision-making trees which is helping us to take the decision. This is the reason why this approach is vastly popular in Machine learning algorithm because it can make complex decision by diving it into simple steps.

Decision trees can solve both classification and regression. For classification the tree will determine a set of logic containing if-then condition to classify the problem and take decision according to that. For example, the discrimination between three different types of fruits based on their certain features like color, shape etc.

Decision trees can also be used for regression. Regression tree is used when the target variable is numerical. Each targeted numerical or continuous variable is labeled as independent variable and fitted in regression model.

Decision tree is very popular because,

a. Simple to visualize, interpret & understand.
b. Huge effort is not needed to prepare data.
c. Linearity or Non-Linear parameters do not affect the performance
d. Categorical or Numerical both can be handled using this.

Decision Tree algorithm have some disadvantages too.

a. Noisy data overfitting occurs
b. When the decision tree is complicated it tends to have low biased. Model become unstable for low biased and it became difficult to work with the data.
c. Decision tends to give high variation where model can get unstable due to small variation of data.

Another important term in decision tree is entropy. Entropy is the technique to measure the randomness unpredictability in a dataset. Assume a dataset of flowers, which contain 7 flowers. 6 of them have different shapes and other attributes. Then the entropy will be high because there is more randomness and unpredictability in the data.

In the next part, we will discuss about Random forest where more than one decision trees are used to classify data-sets.

3.3.4 Random Forest

Random forest method operated by lots of decision trees. First of all, multiple decision tree has been constructed in the training phase. Random forest takes the decision based on the selection of the data. Random forest chooses the decision based on majority of trees decision. As the forest made of huge number of trees, the random forest also follows the same principle with lots of decision trees.

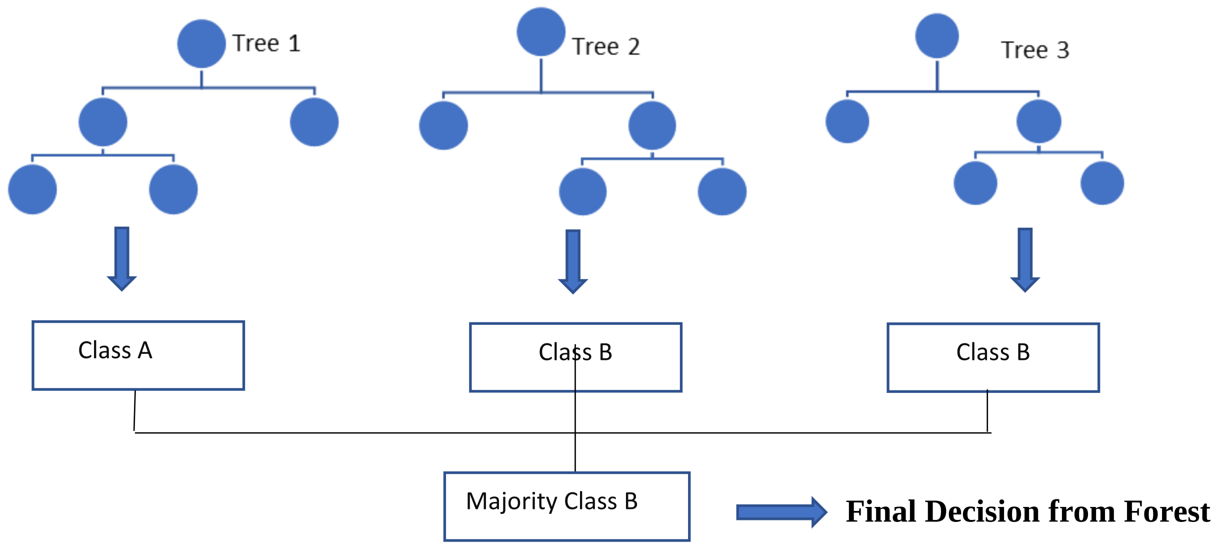


Figure 3.5: Random Forest

In the above figure portrays we are taking three decision trees instead of one tree & making it look like a forest of trees. From the above picture it has three different trees with different decision branches which is built on different attributes. If we want to predict the data through this random forest it will compare the test case with different trained decision trees. The first tree is giving the test data is from Class A whereas tree two and three both are suggesting its from Class B. So, taking the majority we can say the data set is from Class B.

Here an important observation is, in the first tree the prediction is wrong. This is problem or limitation of the decision trees which can be easily solved by random forest. Because it predicts based on several decision trees combined.

The advantages of Random forest are,

a. No overfitting because of the number of multiple trees.
b. Less Training time is required.
c. High accuracy is obtained for large data also run efficiently on large database.
d. Random forest is very useful when there is missing data.

Besides having huge amount of advantages, there are very few disadvantages too. The first disadvantages in random forest is, the number of trees has to be chosen. Also, in random forest there are no interpretability.

The application field of Random forest is quite big.
In the figure below, we can see moving cars and buses. Random forest is used for object detection, where a dataset about objects are given. There are certain number of trees giving the information about object color, shape, model, width, height etc. So, when this object is moving, the accuracy will be lower.

Figure 3.6: Object Detection in Random Forest [5]

Random forest can also be used in game console. Where it traces the moving body parts and adjust the game according to it. Random forest can also be used in medical diagnose where body parts can be traced for medical analysis. Random forest can also be used in remote sensing. Used in ETM devices to collect images of the earth surface.

3.3.5 Naïve Bayes

Have we thought how our emails provider services spam filtering service and filters the unwanted emails in a regular basis or how online news channel does their text classification? The answer of this question is Naïve Bayes classifier.

Naïve Bayes Classifier is an efficient machine learning approach which uses the Bayes theorem for predictive analysis. The principal of the Naïve Bayes works on the conditional probability given by the Bayes theorem. Naïve Bayes algorithm is simple but powerful algorithm prediction.

To understand Naïve Bayes theorem, we need to know basic Bayes theorem,

Bayes theorem gives the conditional probability of an event A given another event is already occurred.

$$\boxed{P(A|B) = \frac{P(B|A)P(A)}{P(B)}}$$

Where,

$P(A \mid B)$ = Conditional probability of A given B

$P(B \mid A)$ = Conditional probability of B given A

P(A) = Probability of event A

P(B) = Probability of event B

A famous coin toss example can be used to describe the Naïve Bayes theorem.

Figure 3.7: Example of Coin Toss

the sample space there are four possible results if we take two coins {HH,HT,TH,TT}

P(Second coin is Head given first is Tail)

P(A|B) = [P (B|A) * P(A)] / P(B)

= [P (First coin is tail given second coin is head) *P (Second Coin Being Head)]/ P (First coin being tail)]

There are three types of Naïve Bayes classifiers,

a. Gaussian
b. Multinomial
c. Bernoulli

Gaussian Naïve Bayes model is used for classification where it assumes that features follow a continuous process or normal distribution. Week or month can be classified with this model because the features are constant and it is not changing.

Multinomial Naïve Bayes model is used for classification assumes that features are discrete. Human brain cell prediction can be done by this model where the attributes are discrete.

Bernoulli Naïve Bayes Model is used for binary features. If the features can be represented by 0 and 1 then Bernoulli Naïve Bayes is useful.

Naïve Bayes classifier is very popular because it lots of advantages.

a. Naïve Bayes can be used for both binary and multinomial problems.
b. Naïve Bayes algorithm requires less training data
c. Naïve Bayes is highly scalable and not sensitive to irrelevant features etc.

The disadvantage of the Naïve Bayes is, it makes a strong assumption on data distribution. Another problem is dependencies exits among variables,
Naïve Bayes Classification in real world application. Spam filtering, Medical Diagnose, Object and face data recognition, weather prediction and news classification.

In conclusion we can say that though Naïve Bayes Algorithm have some disadvantages but for its simple and easy to implement nature it is a useful classifier model for many applications.

In the next part K Nearest Neighbor has been discussed.

3.3.6 K Nearest Neighbor Algorithm

K Nearest Neighbor Algorithm (KNN) is a very simple and essential way to classify data. It is one of the supervised machine learning algorithms. It is used to solve classification and regression related problems. The very basic way to describe K Nearest Neighbor Algorithm is, it stores or memorize from all the available observation or available cases labeled as test set. When a new data is given, it classifies the new data cases or unlabeled observation with labeled observation based on a similarity measurement. KNN makes prediction based on how similar training sets are to the new. The more similar value will be classified as same label. K Nearest Neighbor Algorithm is supervised. KNN is a one kind of instance based or lazy learning.

K Nearest Neighbor Algorithm is a supervised learning algorithm which is a context of machine learning or artificial intelligence. In supervised learning, both input and also some desired output is given & based on the given information it predicts future values.

K Nearest Neighbor Algorithm is a lazy learned, which is exactly opposite of eager learning. Here the training data set does not execute or delayed until a new query is made to the system. It is called lazy because it does not create model instances in advance. But in eager learning it tries to generalize the training data before receiving new queries. The advantage of the lazy learner is, here the targeted function will be approximated locally. So, it can solve multiple problem simultaneously.

The problem with the lazy learning is it takes a huge memory space to store the training dataset, also here noisy data case increases because no abstraction made during the training of the datasets. This lazy learner is very useful when dataset is big with less attributes

KNN algorithm working procedure, K is the number of nearest neighbors. If we consider three classes X and Y where X is Square Shapes and Y is triangular shapes (both are trained dataset) and Z is Rectangular which is the test data.

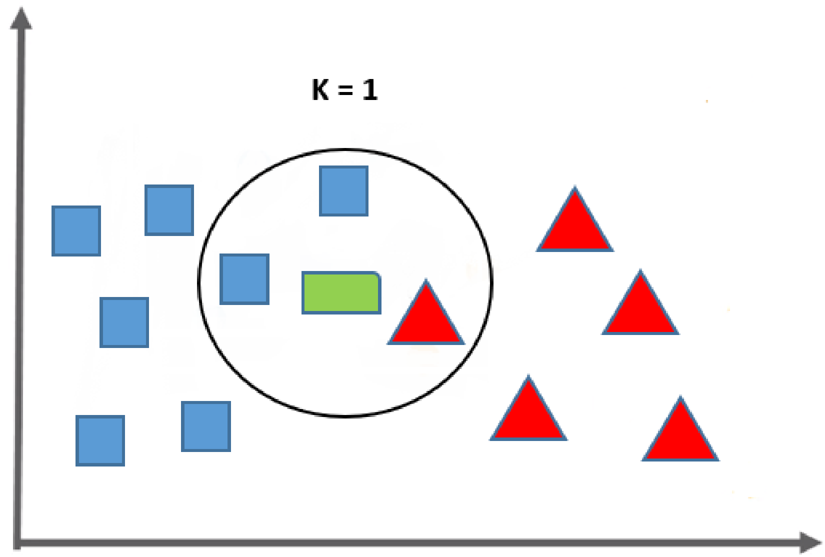


Figure 3.8: K Nearest Neighbor Algorithm

Here we are considering K nearest neighbor is 1. Here X is labeled as blue box and Y is
A new data Z is labeled as green box. In K =1 we have to predict the Z dataset similarity based on our evidence from the trained model. When K = 1, we can see two square boxes and one triangular.

Majority is square, it is nearest to X than Y. So, this will be predicted as X not Y. If the value of K increases a certain amount, the accuracy of this classifier will increase
KNN algorithm uses the least distance measure in order to find its nearest neighbors. To calculate the distance there are several distance measurement ca be used. The most popular are Euclidian distance and Manhattan distance.

To calculate the Euclidian distance, the distance between two points in a plane with coordinates (x, y) and (a, b) is,

$$dist((x, y), (a, b)) = \sqrt{(x - a)^2 + (y - b)^2}$$

To calculate the Manhattan distance,

Manhattan distance between x and y is,

$$|a-c|+|b-d|.$$

Where X = (a,b) and Y = (c,d)

After calculating the distance measurement KNN gives a prediction for the test data.

The value of K should be a trade-off between time vs accuracy. Generally, K = 5 has more accuracy than K = 1. Because it takes more dataset to compare and the chances of finding similar will increase. But when K = 30, the accuracy can be decreased because the local region becomes big and it’s very tough to find the similar neighbors.

Choosing the exact value of K is a bit tricky. When the value of K is small, then noise can have high influence on the data set. Whereas if the value of K is very big then computation cost will be huge. It will take more memory spaces and also the accuracy rate will be affected.

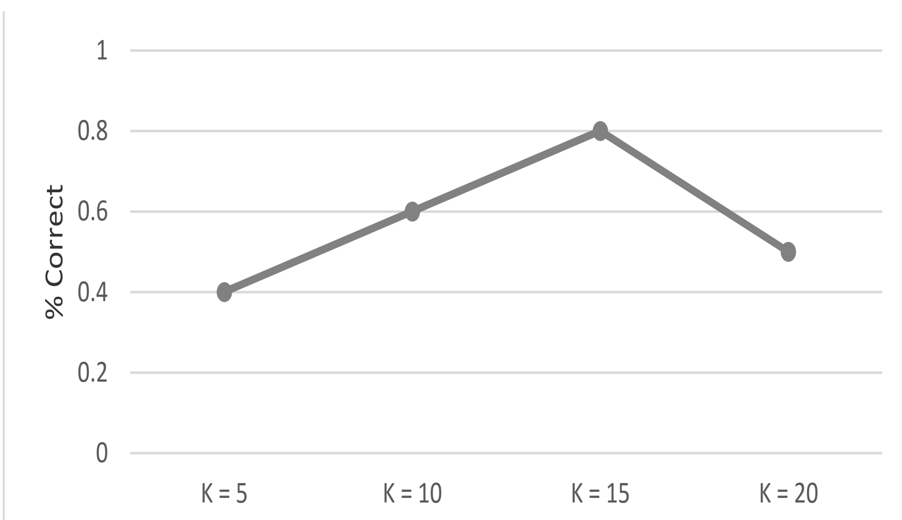


Figure 3.9: Behavior Classification using KNN

In this figure we can clearly see how behavior of the classification changing. For a small value of K, the accuracy is lower also when the values of K are increasing after a certain

number it the accuracy is also decreasing. By trading off the values, we get the desired K value where accuracy is good and also it gives lower computation cost and time.
KNN is hugely popular because it is very simple to implement. Amazon Recommended system uses this algorithm. It is also used in text mining and different types of medical, agriculture related applications.

In conclusion we can say though it has a problem because it takes huge memory but the advantage of KNN algorithm is, it's very effective when training data set is large. KNN is robust to the noisy training data and also useful for non-linear data because no assumptions about the data.

Table 3.2: Comparison of Different Classifiers

Classifier	**Advantages**	**Disadvantages**
KNN	Simple, building model is cheap	Classification is slow
Naïve Bayes	Requires less amount of training data	Dependency exits among variable
Decision Tree	Easy to visualize & understand	Overfitting
Random Forest	Error rate is low, Good for missing data reorganization	Calculation can get very complex
Logistic Regression	Give good information about features	Fall short if decision boundary is not linear
SVM	Very Few irrelevant features	Training is time consuming

3.3.7 Comparison Factors

Before we discuss the comparison factors, accuracy, precision, recall and f1 score, first we have to discuss four basic result types.

a. True-positive: When we are feeding known data to the model to see if it gives correct result or not, if a data is known to be true, and the model also detects it as positive/true, then that case is considering true-positive.
b. False-negative: If a data is known to be true but the algorithm yields a negative result, that case if considered true-negative.
c. False-positive: If a data is known to be false but the algorithm terms it as true, then that case is taken as false-positive.
d. True-negative: When a data is known to be false and the algorithm tells it as a negative data as well, then we term that case as false-negative.

Here is the figure,

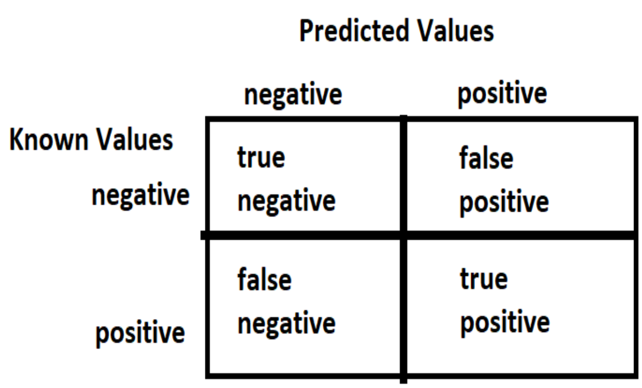


Figure 3.10: Confusion Matrix

In the next part we will discuss about the four parameters by which we have measured and compared our datasets.

3.3.7.1 Accuracy

Accuracy denotes the closeness of our result value to a standard known value. For example, if we are measuring an objects weight, and we know it already that the weight is 10 units, but the system is showing it to be 4 units, then the system is not accurate.

3.3.7.2 Precision

Precision means the closeness of measured results from two or more measurements. For example, if we measure the weight of an object 10 times, and every time the result is 5 units, then the measurement can be called very precise.

It is important to notice here is that system's measurements, to be called precise, doesn't necessarily have to be accurate. If all of the 10 measurements give inaccurate result, but all of them are close to each other, then, although they are inaccurate, they will have a high precision.

On the other hand, accuracy is also independent of precision. If a measurement is accurate, meaning it measures the item correctly, or close to the correct known value, but the measurements are not close to each other, then the measurements, although they are accurate or of high accuracy, will not have precision.

Precision can be calculated as the following

$$\textbf{Precision} = \frac{\textbf{True Positive}}{\textbf{True Positive + False Positive}}$$

Figure 3.11: Equation of Precision

Precision is used when the cost of getting a false positive is high. Like in spam detection in emails. If an email, which is not spam is calculated as spam, then we might get an important mail put in the spam box. Here precision would be important.

3.3.7.3 Recall

Recall can be calculated as the following:

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

Figure 3.12: Equation of Recall

Recall is needed when the cost of getting a false negative is high. For example, in fraud detection. If an activity is actually fraud, meaning positive, and the system detects it as not-fraud or negative, then we might get some potential danger approved by our system. So, we need to use recall here.

3.3.7.4 F1 Score

F1 score is a function of precision and recall we use F1 score when we need a balance between precision and recall.

$$F1 = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

Figure 3.13: Equation of F1 Score

In this thesis, we will compare the machine learning classification algorithms in our focus with respect to the following measures.

a. Accuracy
b. Precision
c. Recall
d. F1 Score

In this chapter we have discussed about different machine learning classifiers. Each of the classifier has its own pro and cons. Based on the application different classifier is used.

CHAPTER FOUR

EXPERIMENTAL ANALYSIS

4.1 Overview

This thesis work compares different machine learning classification algorithms' performance in terms of detecting fake news. We are using parameters like accuracy, precision, recall and F1 score to compare the algorithms. In chapter 3, we have discussed different machine learning classification algorithms that we will be using in terms of theory and how they work internally. In this chapter we analyze them in practical setup. We have written our code in python and implemented the tf-idf and classifier algorithms from the sklearn library. In this chapter, first, in section 4.2 we describe the setup of the system where we are going to compare the algorithms. Then we will discuss the performance of each of the algorithms in section 4.3. Finally, in section 4.4 we will compare the performance parameters of all those algorithms side by side.

4.2 System Setup

In this section, we are going to describe the system setup that we have used to run the code and get the results for comparison.

Table 4.1: Computer Specification

Processor	Intel® Core™ i5-8300H
Generation	8^{th}
Clock Speed	2.3 GHz
Max Turbo Frequency	4.0 GHz
Number of Cores	4
Number of Threads	8
Cache	8MB

Table 4.2: Memory Specification

Physical Memory	8GB
Physical Memory Type	4
GPU	4GB
GPU Type	NVIDIA GeForce GTX 1050Ti

Table 4.3: Software Specification

Software Name	Type	Version	Architecture
Anaconda	Python distributor	5.3.1 Python 3.6.5	64bit (x68)
Spyder	Python IDE	3.2.8 Python 3.6.5	64bit (x68)

Table 4.4: Window Specification

Name	Windows 10 Home
Version	10.0.17134
Build Number	17134
System Type	64 bit

4.3 Performance on different classifiers

In this section we will see performance of different classifiers based on the comparison factors.

4.3.1 Performance of Logistic Regression

We got the following confusion matrix for logistic regression

	0	1
0	103	1
1	18	19

Figure 4.1: Confusion Matrix of Logistic Regression

Measurements we got from our code is as following:

Measurements using Logistic Regression:

Accuracy: 0.8652482269503546

Precision: 0.950

Recall: 0.514

F1: 0.667

As we can see, Logistic regression has a fairly low recall and f1 score.

4.3.2 Performance of Support Vector Machine

We got the following confusion matrix for SVM in our code:

	0	1
0	97	7
1	11	26

Figure 4.2: Confusion Matrix of SVM

Our measured result for SVM is as following:

Measurements using SVM:

Accuracy: 0.8806818181818182

Precision: 0.829

Recall: 0.708

F1: 0.764

As we can see, SVM has a higher Recall, F1 and accuracy than logistic regression and a lower precision.

4.3.3 Performance of Decision Tree

We got the following confusion matrix applying decision tree classification algorithm:

	0	1
0	97	7
1	13	24

Figure 4.3: Confusion Matrix of Decision Tree

We got the following measurements in this case:

Measurements using Decision Tree:

Accuracy: 0.8297872340425532

Precision: 0.697

Recall: 0.622

F1: 0.657

As we can see, we have already found classification algorithms with better measurements in all four of these areas than decision tree.

4.3.4 Performance of Random Forest

Our confusion matrix for random forest classification was as following:

	0	1
0	102	2
1	15	22

Figure 4.4: Confusion Matrix of Random Forest

The measurements we got from this algorithm are as follows:

Measurements using Random Forest:

Accuracy: 0.8636363636363636

Precision: 0.900

Recall: 0.562

F1: 0.692

As we can see, random forest's recall value is even lower than logistic regression. It has a lower accuracy and f1 score as well.

4.3.5 Performance of K Nearest Algorithm

We applied k-nearest neighbor or KNN and got the following confusion matrix:

	0	1
0	94	10
1	11	26

Figure 4.5: Confusion Matrix of KNN

We got the following measurements:

Measurements using KNN:

Accuracy: 0.8226950354609929

Precision: 0.620

Recall: 0.838

F1: 0.713

As we can see, KNN has a high Recall. But the accuracy and precision score is lower than logistic regression, random forest and SVM.

4.3.6 Performance of Naïve Bayes (Bernoulli)

The following confusion matrix was found when we applied Bernoulli algorithm:

	0	1
0	86	18
1	10	27

Figure 4.6: Confusion Matrix of Naïve Bayes

The measurements we got from this classification algorithm is as follows:

Measurements using Bernoulli:

Accuracy: 0.8014184397163121

Precision: 0.600

Recall: 0.730

F1: 0.659

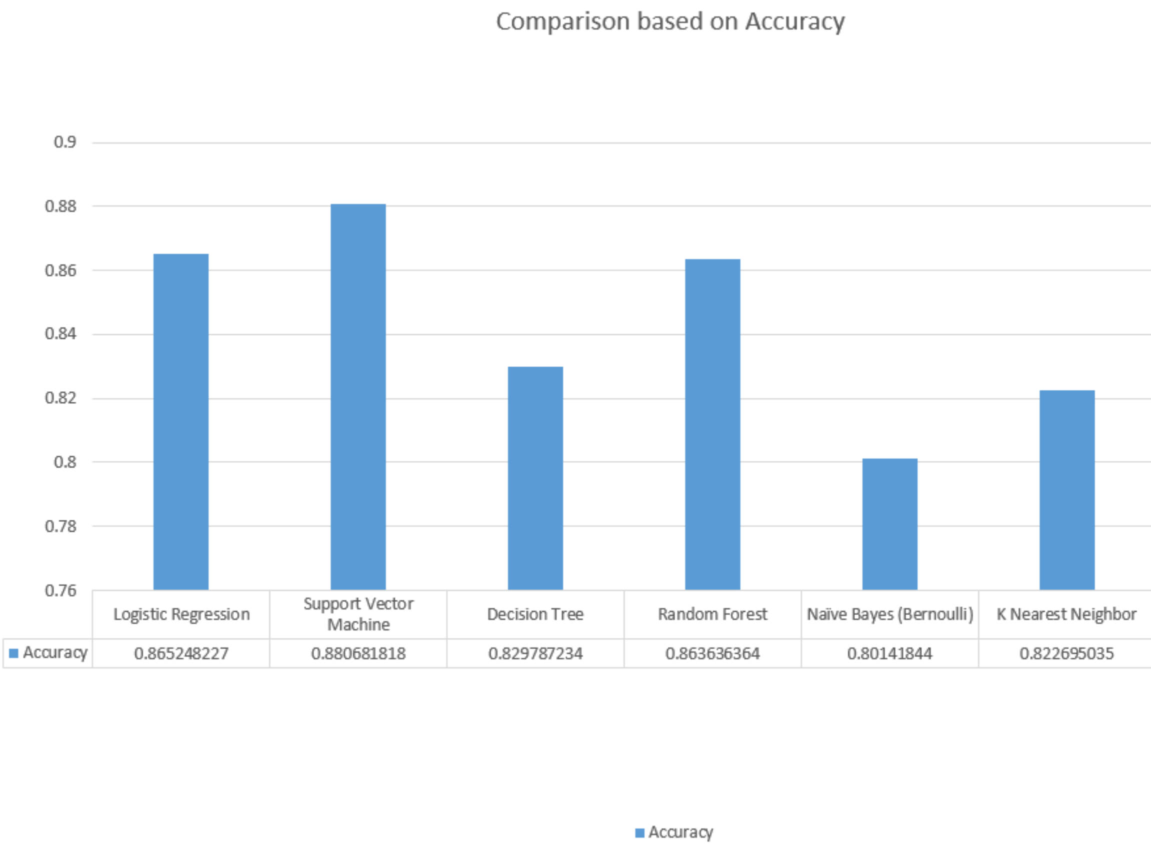
We can see, Bernoulli has a lower Accuracy, Precision and F1 score.

4.4 Comparison

In the previous section, we measured the accuracy, precision, recall and f1 score of various machine learning classification algorithm. In this section, we are going to compare the algorithm based on these measurements.

4.4.1 Comparison based on accuracy

We plot the accuracy measurements of the 7 different classifier algorithms and got the following chart in result:



	Logistic Regression	Support Vector Machine	Decision Tree	Random Forest	Naïve Bayes (Bernoulli)	K Nearest Neighbor
Accuracy	0.865248227	0.880681818	0.829787234	0.863636364	0.80141844	0.822695035

Figure 4.7: Comparison based on Accuracy

As we can see from charts, SVM provides the highest accuracy, while Naïve Bayes (Bernoulli) provides the lowest accuracy for our dataset

4.4.2 Comparison based on Precision

We put the precision of the algorithms that we measured and got the following chart showing us the results:

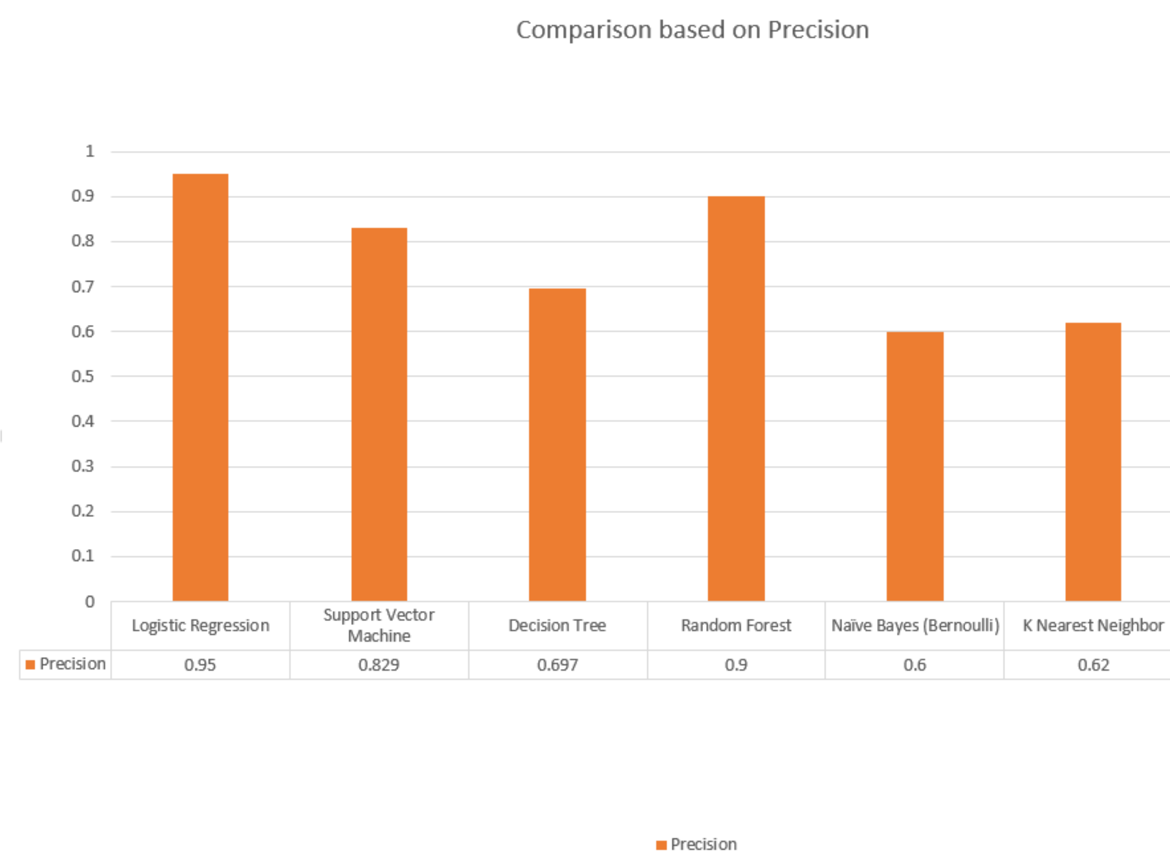


Figure 4.8: Comparison based on Precision

As we can see from the chart,

Logistic regression has the highest precision testing our dataset, followed by random forest. On the other hand, Naïve Bayes (Bernoulli) has the lowest precision, while KNN is the second from the bottom.

We measured the recall values of the 7 different classification algorithms for our dataset and got this chart from the results.

4.4.3 Comparison based on Recall

Figure 4.9: Comparison based on Recall

As we can see from the chart, K Nearest Neighbor has the highest recall value, followed by Neural Networking. On the other hand we can see that Logistic regression has the lowest recall value for our data set while Random Forest has the second lowest one.

4.4.4 Comparison based on F1 Score

We measured the f1 score of all the different classifier algorithms we used and plotted them to get the following chart:

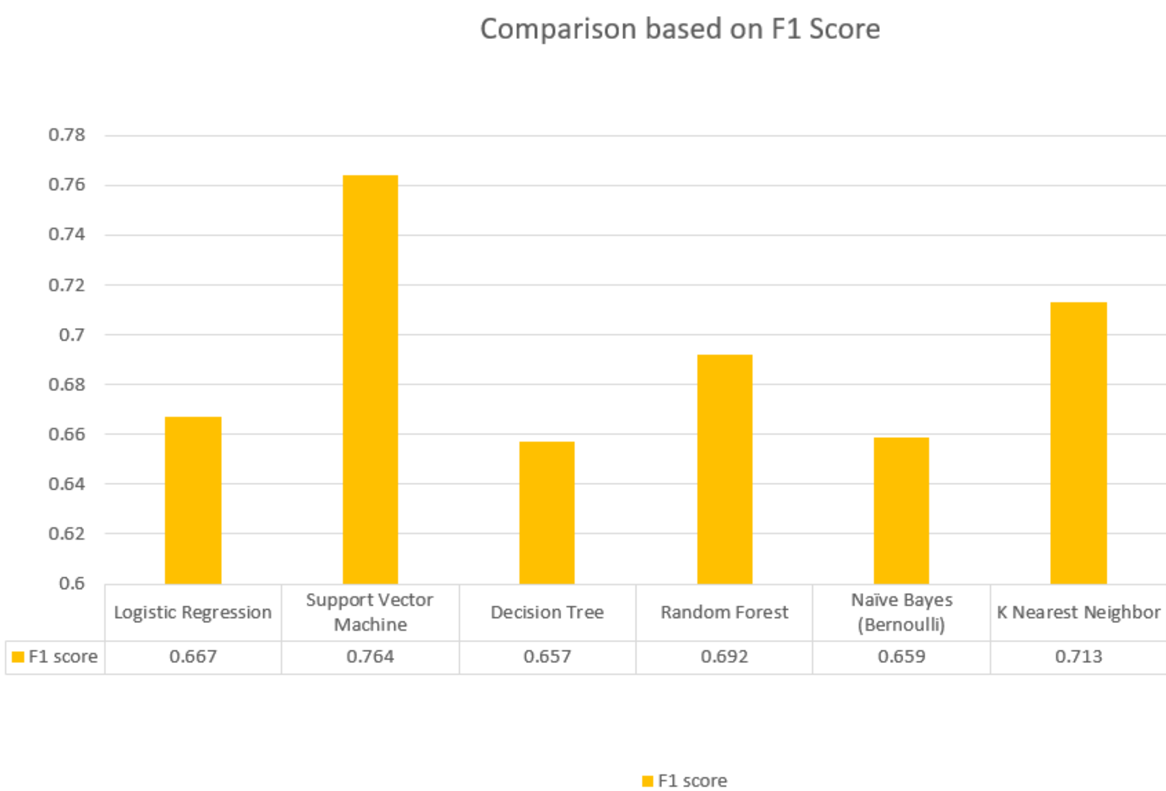


Figure 4.10: Comparison based on f1 score

As we can see from the chart, Support Vector Machine has the highest F1 Score for our dataset followed by Neural Networking. On the other side, Decision Tree has the lowest f1 score for our dataset while Naïve Bayes (Bernoulli) has the second lowest f1 score.

4.4.5 Overall Comparison

Plotting all the four factors we are considering, accuracy, precision, recall and f1 score in one chart, we get the following:

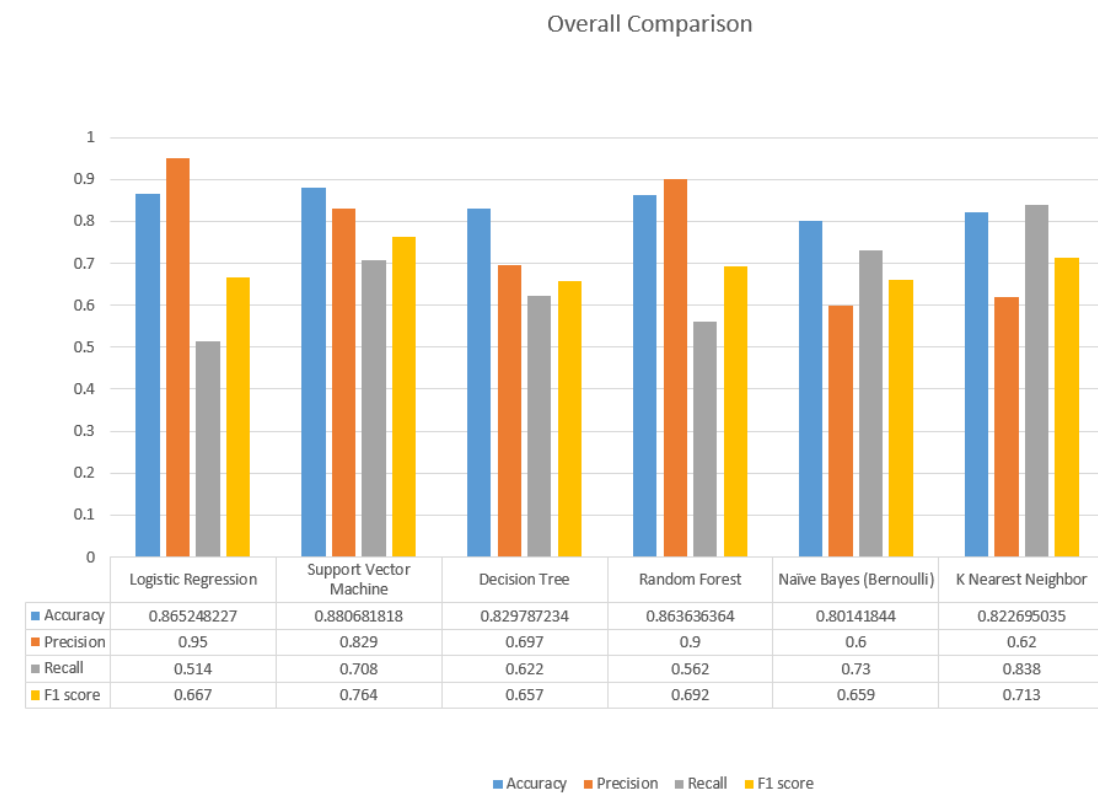


Figure 4.10: Overall Comparison

The following table shows all the 4 different comparison factors of the 7 different machine learning classifier algorithms in a tabulature format

Table 4.5: CPU Specification

Algotithm Name	Accuracy	Precision	Recall	F1 score
Logistic Regression	0.865248227	0.95	0.514	0.667
Support Vector Machine	0.880681818	0.829	0.708	0.764
Decision Tree	0.829787234	0.697	0.622	0.657
Random Forest	0.863636364	0.9	0.562	0.692
Naïve Bayes (Bernoulli)	0.80141844	0.6	0.73	0.659
K Nearest Neighbor	0.822695035	0.62	0.838	0.713

In this chapter, we discuss about the difficulties we faced, future works and our concluding remarks. So, this chapter discusses our experience doing the work and also our future plan in this road further.

CHAPTER FIVE

CONCLUSION

5.1 Difficulties

There were many difficulties that we had faced while working on this paper. The first one was the unviability of training data. We could only manage a dataset of 701 news. Of this dataset, 498 were fake news and 203 were real news. So, there was a shortage of real news while an abundance of fake news. We feel a dataset, which was high in number and also balanced in terms of the number of real and fake news would help us to train better. With our little sized dataset, we faced difficulties implementing the classification.

5.2 Future Work

There is always a room for improvement for any research. This one is not an exception to that either. We have found some areas that can be worked on in the future.

a. More classification: We have classified the news as only real or fake. But a news can have other classifications, like opinion, miss interpretation, satire and others. It would bring more accuracy to the system if we categorize a news as more of those classes other than just black and white (real or fake in this case)
b. Other algorithms: There is a possibility to find better accuracy, precision, recall and f1 in algorithms that were not used here.
c. Image implementation: We have tried to detect fake news based on content of the news. But we have only tested it with the text of the news. Sometimes, fake news can be spread with fabricated and edited images. So, implementing image processing in detecting fake news is also an area of improvement.

5.3 Concluding Remarks

In this paper, we have tried to compare 7 different machine learning classification algorithms based on four different comparison factors, accuracy, precision, recall and f1 score in detecting fake news. First, we took a dataset of 701 news categorized as either fake or real. Then split the data, 80% of it is used for training and 20% for testing. We feed the training data to 7 different machine learning classification algorithms to calculate the four comparison factors. Finally, we noted and compared the result. We have found that Support Vector Machine has the highest accuracy while Naïve Bayes (Bernoulli) has the lowest. In terms of precision, Logistic Regression is the highest, while Naïve Bayes (Bernoulli) is the lowest. While measuring the recall value, K Nearest Neighbor had the highest values, while Logistic Regression had the lowest value. Finally, Support Vector Machine had the highest f1 score and Decision Tree had the lowest.

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